



Full Length Article

Design of an accurate end-of-arm force display system based on wearable arm gesture sensors and EMG sensors



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ABSTRACT

Most upper limb rehabilitation patients are still hard to feel the accuracy force they have imposed in the end of arm after a systematic upper limb rehabilitation. In order to provide an accurate end-of-arm force for those disabled people, a force display system based on wearable arm gesture sensors and Electromyographic (EMG) sensors is designed and given in this paper. The wearable arm gesture sensors and EMG sensors are specially placed to detect the arm gesture and the EMG signal of the arm, and a force sensor is used to measure the force and verify the force display effect. In order to control the rehabilitation arm move slowly at a constant speed, the kinematic model of the upper arm is analyzed. A special experiment platform is established so as to get the simultaneous data of end-of-arm force and the arm gesture and EMG signal, then the Generalized Regression Neural Network (GRNN) is brought in to catch the relationship between them. A group of horizontal movement experiment and vertical movement experiment are designed specially and verify the feasibility and effectiveness of the system. The result shows that the information fusion based on GRNN for this system could accurately display the end-of-arm force.

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1. Introduction

Nowadays, many disabled people and old people have a dyskinesias problem with their arms caused by stroke or an unfortunate accident [1,2]. As the improvement of robot skill and medical technology, many effective therapies [3,4] are brought in to try to solve the arms' dyskinesias problem, especially the special rehabilitation training for the disabled arm, and many rehabilitation robots [5] with different structure and function have been designed and took into the application.

Colombo [6] presented two rehabilitation robot devices for the upper limb, including a one degree of freedom wrist manipulator and a two degree of freedom (DOF) elbow-shoulder manipulator, and an admittance control strategy is given to control the robot move smoothly with a mean velocity. Chen [7] presented an assistive control system with a special kinematic structure of an upper limb rehabilitation robot based on the NTUH-ARM. Three rehabil-

itation modes including active mode, assistive mode and passive mode are conducted to help the impaired limb move and do the rehabilitation exercise. Xu [8] presented control strategies for an upper-limb rehabilitation robot based on the BARRET WAM, including a fuzzy-based PD position control strategy under the passive recovery exercise to control the WAM Arm stably and smoothly to stretch the impaired limb to move along predefined trajectories, and a fuzzy logic regulator in the active motion mode to adjust the desired impedance between the robot and impaired limb to generate adaptive force in agreement with the change of the impaired limb's muscle strength.

Although some of the upper limb rehabilitation robot could positively make the arms recovered with much flexibility and controllability, however, the arms are still impossible to be healed as a regular arm of a normal person, and the new problem which always occurs after the arm upper limb rehabilitation was completed is that the rehabilitation patients are still hard to accurately feel how much force they have used in the end of the impaired arm, and they could not almost handle an object stably. Therefore, in order to solve the force display problem for the rehabilitation patients, an accurate end-of-arm force display system is designed in this paper based on information fusion with arm gesture and

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Fig. 1. Body sensors network kit.

EMG signal. A lot of information fusion approaches [9–11] have been presented by the researchers all over the world, and some of them are put into application and works well. In this paper, GRNN is chosen to fuse with arm gesture and EMG signals because it has a great advantage in nonlinear mapping ability and learning rate.

The remainder of this paper is organized as follows. First, Section 2 describes the system architecture, including the wearable arm gesture sensors part, wearable EMG sensors part and the force measurement and display part. Kinematic model of the upper limb is established in Section 3, and information fusion approach based on the GRNN is described in Section 4. In Section 5, experiments platform is specially designed and described, and the comparison between the force derived by the force display system and the real force is given, and at last conclusion is given in Section 6.

2. System architecture and analysis

In order to provide an accurate end-of-arm force for the disabled people, especially for those who have completed upper limb rehabilitation but still could not have the accurate sense of force in the end of arm, a force display system is established, including the wearable arm gesture sensors part, wearable EMG sensors part and the force measurement and display part.

2.1. Wearable arm gesture sensors part

Wearable arm gesture sensors part is constituted of an arm sleeve, a short glove, and three-dimension (3D) angle sensors, and other auxiliary components, as shown in Fig. 1. The arm sleeve and the short glove are designed as an open circle and could be easily fastened, and it is portable and suitable almost for every adult people. The 3D angle sensors are placed on the different position individually to detect the angles of different part of the arm, as shown in Fig. 2. Therefore, it is easy to get the eulerian angles of the upper arm, forearm and the hand.

The core device of the 3D sensor is ADXL362(Analog Device). It's ultralow power, 3-axis MEMS accelerometer. It can provides 12-bit output resolution. Measurement ranges of ± 2 g, ± 4 g, and ± 8 g are available, with a resolution of 1 mg/LSB on the ± 2 g range.

2.2. EMG sensors part

EMG signal is a compositive superposition of many muscle motor unit action potentials from both time and space domain, and reflect the status of neuromuscular function. EMG signal is a kind

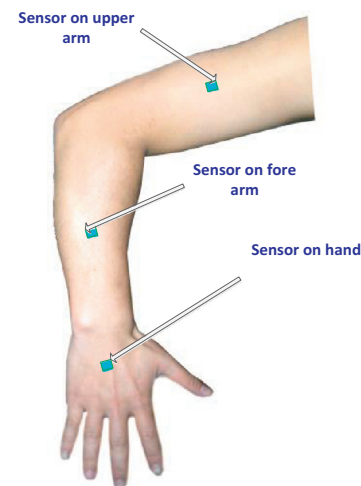


Fig. 2. Arm gesture sensors distribution.

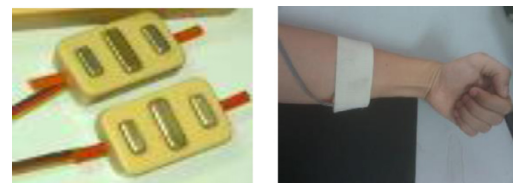


Fig. 3. EMG sensors.

of quite weak signal with range from 100 to 5000 μ V. Every EMG sensor has three poles, including '+', 'GND' and '-', and the difference of every two near poles is 6 mm, as shown in Fig. 3.

The structure diagram of the conditioning circuit of the EMG sensor is shown in Fig. 4. The voltage gains of the two amplifiers, A1 and A2, are set as 15 and 40 respectively. The high-pass filter (cut-off frequency: 10 Hz) is used to remove the direct current component from the signal. The low-pass filter (cut-off frequency: 500 Hz) is used to eliminate the high-frequency noises. The voltage gains of these two filters are about 1.3. The notch filter is used to reduce the 50 Hz power-line interference. Since the input range of the data collector is from 0 to 3 V, we design the level-rising circuit to make the voltage of the signals greater than 0 V.

A group of two EMG sensors are attached and clinged to the skin of fore arm, and have enough contact area with fore arm. EMG signal captured by the EMG sensors is a time series signal which can describe the characteristics of the hand action after necessary

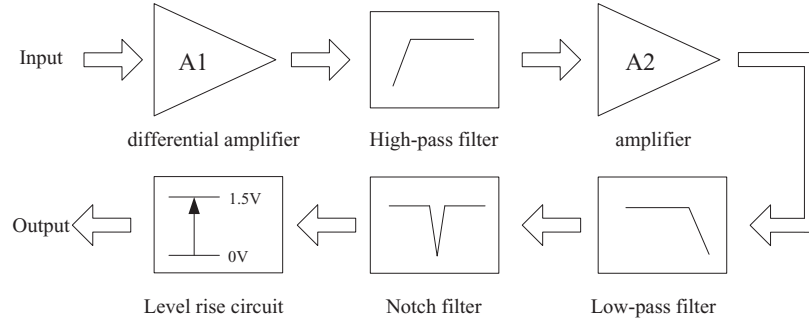
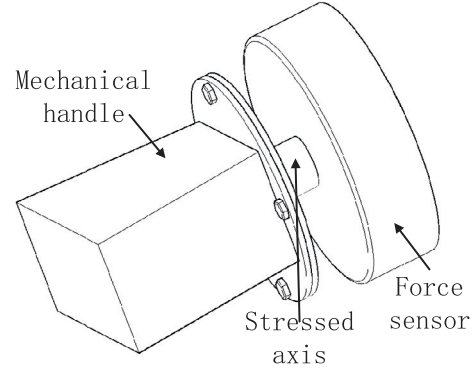


Fig. 4. The structure diagram of the conditioning circuit.



Fig. 5. Force sensor.



preprocessing and feature extraction. Here we use time domain feature extraction methods. The usual time domain feature extraction methods are mean absolute value (MAV), variance (VAR), zero crossings (ZC) and Willison Amplitude (WA). The mathematical equation of these two features are as follow:

$$MAV_i = \frac{1}{N} \sum_{j=i-N+1}^i |x_j| \quad (1)$$

$$VAR_i = \frac{1}{N} \sum_{j=i-N+1}^i x_j^2 \quad (2)$$

where, x_j is the j th sample data, N is the length of the sliding window.

2.3. Force measurement and display part

The force sensor (measurement range: -30 N – 30 N , accuracy: 2%FS) which we made in our previous work [12,13], is used to measure and verify the value of force, as shown in Fig. 5.

A group of six small vibration motor is designed and placed around the upper arm, as shown in Fig. 6. After the accurate value of the force has been detected and calculated, the vibration motors will provide vibration to remind the user how much force he has imposed.

In the force displaying, the force measured by the sensor is used to modulate the vibration strength (VS) according to the principle as shown in formula (3), and the vibration frequency is set to 100 Hz.

$$VS = \begin{cases} \text{I} & 0 < r \leq 0.2 \\ \text{II} & 0.2 < r \leq 0.4 \\ \text{III} & 0.4 < r \leq 0.6 \\ \text{IV} & 0.6 < r \leq 0.8 \\ \text{V} & 0.8 < r \leq 1.0 \end{cases} \quad (3)$$

where, I, II, III, IV, V are the vibration strength levels, the relationship between vibration strength levels and the voltage applied to

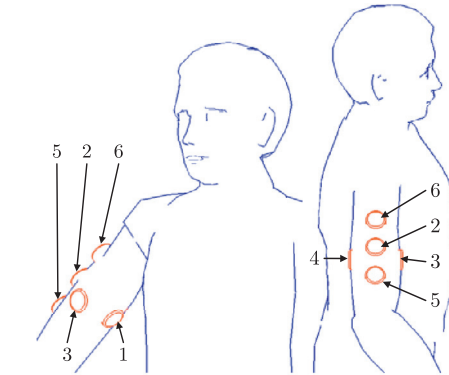


Fig. 6. Vibration motor distribution.

Table 1

Relationship between vibration strength level and voltage applied to motor.

Vibration strength level	Voltage applied to motor(V)
I	1.0
II	2.0
III	3.0
IV	4.0
V	5.0

the motor is as shown in Table 1. r is the ratio of grasp force detected by the force sensors to the maximum allowable value of the grasp force.

$$r = \frac{F}{F_{\max}} \quad (4)$$

where F is the grasp force detected by the force sensors, and $F_{\max}$ is the maximum allowable value of the force.

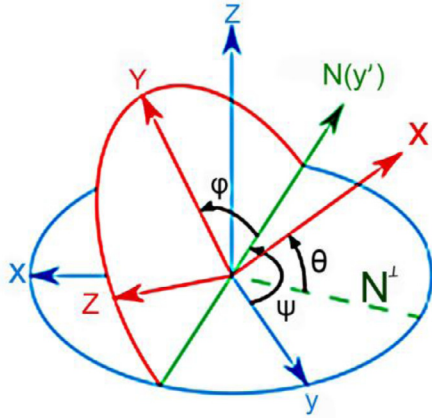


Fig. 7. Quaternions coordinate.

3. Kinematic model of the upper limb

Quaternion is used to describe the rotation angle of every part on the arm, as shown in Fig. 7. Any quaternion $q = [w \ x \ y \ z]$ could be represented as $q = a + bk + cj + di$, where a, b, c, d are real number, and i, j, k are limited by $i^2 = j^2 = k^2 = -1$ and $i^*0 = j^*0 = k^*0 = 1$. Suppose one of arm gesture sensor shows that the eulerian angle ψ, θ, φ is the revolution angle around Z axis, Y axis and X axis respectively. Then the $q = [w \ x \ y \ z]$ could be derived as follows:

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos(\varphi/2)\cos(\theta/2)\cos(\psi/2) + \sin(\varphi/2)\sin(\theta/2)\sin(\psi/2) \\ \sin(\varphi/2)\cos(\theta/2)\cos(\psi/2) - \cos(\varphi/2)\sin(\theta/2)\sin(\psi/2) \\ \cos(\varphi/2)\sin(\theta/2)\cos(\psi/2) + \sin(\varphi/2)\cos(\theta/2)\sin(\psi/2) \\ \cos(\varphi/2)\cos(\theta/2)\sin(\psi/2) - \sin(\varphi/2)\sin(\theta/2)\cos(\psi/2) \end{bmatrix} \quad (5)$$

If the rotation angle of upper arm is q_1 , and the rotation angle of fore arm is q_2 , and the rotation angle of hand is q_3 , the rotation angle of elbow q_a and wrist q_b could be determined by:

$$\begin{aligned} q_a &= q_1^{-1} \times q_2 \\ q_b &= q_2^{-1} \times q_3. \end{aligned} \quad (6)$$

Then the kinematic model of arm could be analyzed, as shown in Fig. 8. The origin point of coordinates O_0, O_1 and O_2 superpose together at the shoulder position, and O_3 is at the elbow position, and O_4 and O_5 superpose together at the wrist position. l_1 and l_2 is the length of upper arm and fore arm respectively. The D-H matrix of a full arm is shown in the Table 2.

Then the coordinate system of the end of arm relative to the coordinate system of the shoulder could be derived:

$$T = A_1 A_2 A_3 A_4 A_5 \quad (7)$$

Where $A_1 = Rot(z_0, 180)Rot(z_1, \theta_1)$

$$= \begin{bmatrix} -\cos\theta_1 & \sin\theta_1 & 0 & 0 \\ -\sin\theta_1 & -\cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (8)$$

$$A_2 = Rot(x_1, -90)Rot(z_2, \theta_2) = \begin{bmatrix} \cos\theta_2 & -\cos\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin\theta_2 & -\cos\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

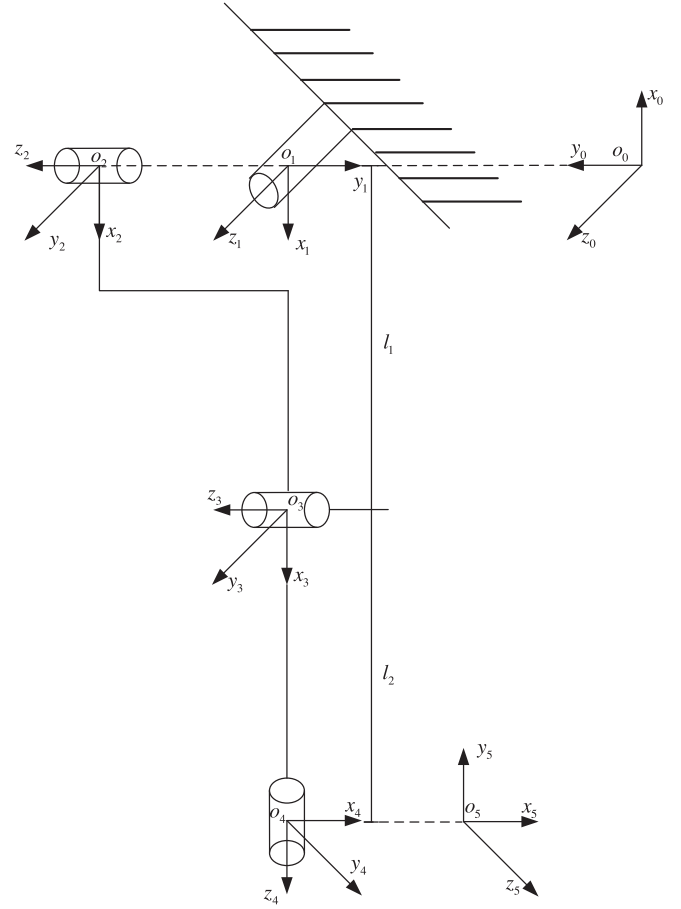


Fig. 8. Kinematic model of arm.

Table 2
DH matrix parameter.

Joint i	Joint angle θ	Angle of distortion α	Linkage length a	Linkage offset d
1	θ_1	180°	0	0
2	θ_2	-90°	0	0
3	θ_3	0°	l_1	0
4	θ_4	90°	l_2	0
5	θ_5	-90°	0	0

$$A_3 = Trans(x_2, l_1)Rot(z_3, \theta_3) = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & l_1 \\ \sin\theta_3 & \cos\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (10)$$

$$\begin{aligned} A_4 &= Trans(x_3, l_2)Rot(y_3, 90)Rot(z_4, \theta_4) \\ &= \begin{bmatrix} \cos\theta_4 & -\sin\theta_4 & 0 & l_2 \\ 0 & 0 & -1 & 0 \\ \sin\theta_4 & \cos\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \quad (11)$$

$$A_5 = Rot(x_4, -90)Rot(z_5, \theta_5) = \begin{bmatrix} \cos\theta_5 & -\sin\theta_5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin\theta_5 & -\cos\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (12)$$

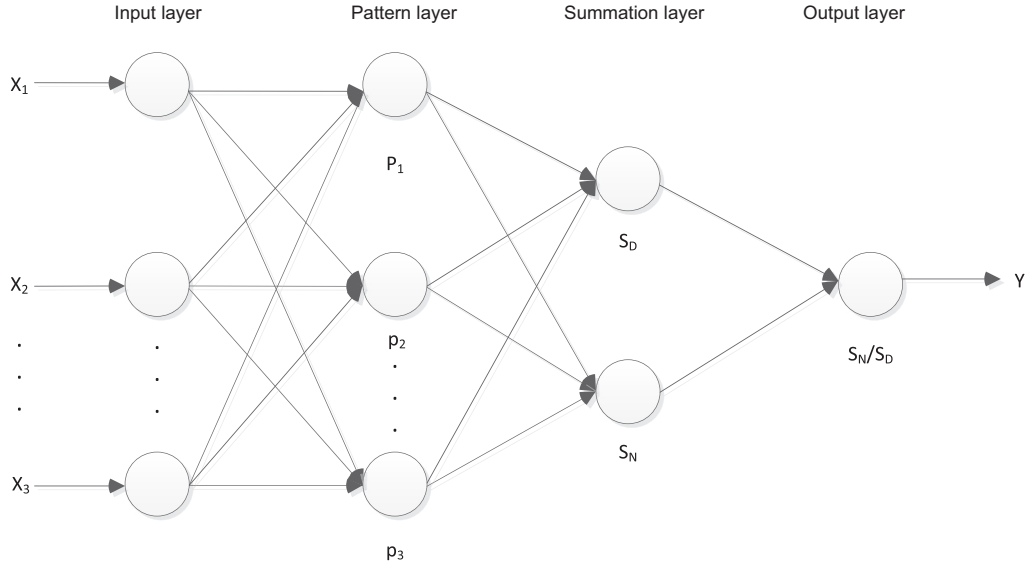


Fig. 9. Schematic diagram of general recurrent neural network.

4. Information fusion based on GRNN

Every single arm gesture with a certain group of EMG signals will produce end-of-arm force with a certain value. Here based on the system established, an information fusion approach with arm gesture sensors and EMG sensors is conducted to fit the value of end-of-arm force.

GRNN is a radial basis function neural network model proposed by Specht in 1991 [14–16]. Even with sparse data in a multidimensional measurement space, the algorithm provides smooth transitions from one observed value to another. Moreover, GRNN has good predictive results for unsteady data.

General regression neural network structure shown in Fig. 9, respectively, include the input layer, pattern layer, the summation layer and output layer.

The number of input neurons is equal to the dimension of the input vector in the learning sample.

Pattern layer neurons correspond to different samples, the transfer function is

$$P_i = \exp \left[-\frac{(x - x_i)^T (x - x_i)}{2\sigma^2} \right], \quad i = 1, 2, 3, \dots, n \quad (13)$$

The summation layer is summed by two approaches. One approach is to sum the outputs of all pattern layer neurons, The weight of the pattern layer and each connection layer is 1, Through this method the calculation formula and the transfer function are

$$\sum_{i=1}^n \exp \left[-\frac{(x - x_i)^T (x - x_i)}{2\sigma^2} \right] \quad (14)$$

$$S_D = \sum_{i=1}^n P_i \quad (15)$$

Another approach is to weight the neurons in all the pattern layers, Through this method the calculation formula and the transfer function are

$$\sum_{i=1}^n y_i \exp \left[-\frac{(x - x_i)^T (x - x_i)}{2\sigma^2} \right] \quad (16)$$

$$S_{Nj} = \sum_{i=1}^n y_{ij} P_i, \quad j = 1, 2, 3, \dots, k \quad (17)$$

The number of neurons in the output layer is equal to the dimension of the output vector, Each neuron divides the output of



Fig. 10. Experiment platform based on WAM manipulator robot.

the summation layer, J-th neuron j-th element output corresponding to the output of y, which is

$$y_j = \frac{S_{Nj}}{S_D}, \quad j = 1, 2, 3, \dots, k \quad (18)$$

After the construction of the GRNN, the 3D position information captured by the 3D sensors and the feature information extracted from the EMG signal are used to train the GRNN.

5. Experiment and analysis

In order to detect the feasibility and effectiveness of the information fusion approach with the wearable arm gesture sensors and EMG sensors, a special group of experiment based on the Barrett WAM arm manipulator robot [17,18] was conducted, as shown in Fig. 10. A handle was specially made and jointed to the force sensor, and then connected to the WAM manipulator robot. As shown in Figs. 11 and 12, experiment candidate are organized to hold the handle and try to make the handle move slowly with a constant speed, horizontally and vertically respectively. The WAM



Fig. 11. Horizontal movement experiment.

manipulator robot was controlled randomly in the active mode and passive mode to make the handle move with different force, and the experiment candidate could feel different force at different time.

Both horizontal movement experiment and vertical movement experiment have the training session and test session. As shown in Fig. 11, during the training session, the horizontal movement was conducted back and forth several times, the data from the arm gesture sensors and the data of the EMG features were as the input of GRNN, and the simultaneous data from the force sensor was sampled as the output of GRNN. After the GRNN training session, it will come to the test session and additional groups of experiment were conducted and used to verify the accuracy of the information fusion approach.

As shown in Fig. 13, which is data of horizontal movement experiment, (a) is data from the arm gesture sensors, (b) is the EMG signals (c) is the end-of-arm force comparison between real value measured by force sensor and the prediction value derived from information fusion based on GRNN.

The vertical movement experiment was conducted similarly with the horizontal movement experiment, and the result was shown in Fig. 14, (a) is data from the arm gesture sensors, (b) is the EMG signals (c) is the end-of-arm force comparison between real value measured by force sensor and the prediction value from GRNN.

As shown in Figs. 13(c) and 14(c), the results show that the force prediction under information fusion based on GRNN with arm gesture sensors and EMG sensors could effectively describe the end-of-arm force accurately, and the prediction curves fit the real value curves well. Then according to the accurate prediction force derived from the information fusion of the body sensors network, the group of six small vibration motor adhered in the arm will produce vibration with the corresponding amplitude to show how much the force is, which means, with the help of wearable force display system, the patient with impaired limb could feel about the end-of-arm force and could be clear about how much force he has imposed in the end of arm.

6. Conclusion

In order to provide an accurate end-of-arm force for the disabled people, especially for those who have completed upper limb rehabilitation but still could not have the accurate sense of force in the end of arm, this paper presented a force display system with



Fig. 12. Vertical movement experiment.

wearable arm gesture sensors and EMG sensors. GRNN is used as the sensor fusion approach to fit the end-of-arm force. After training process, the verification in the test process shows that the prediction value of end-of-arm force via trained GRNN is accurate and feasible. With the help of wearable force display system, the patient with impaired limb could feel about the end-of-arm force and be clear about how much force he has imposed in the end of arm.

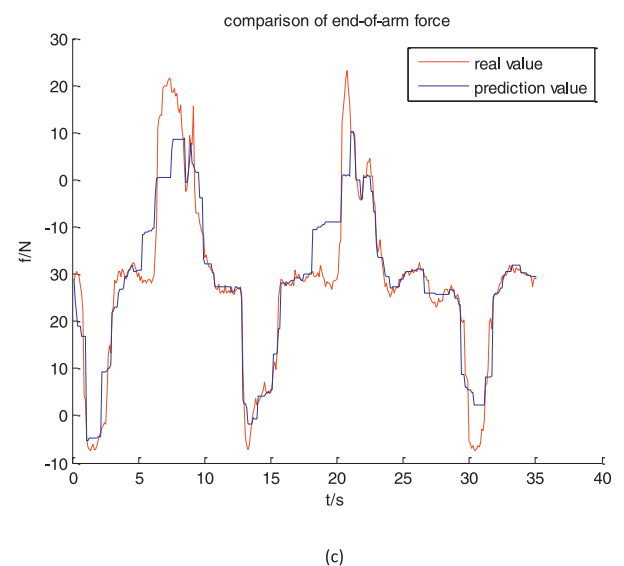
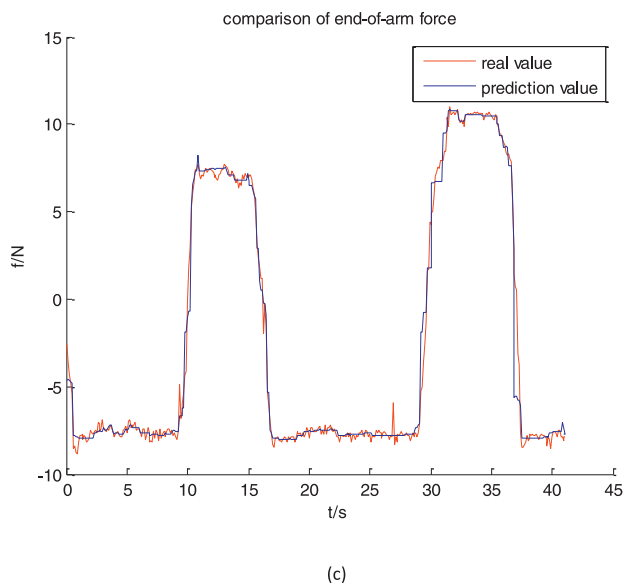
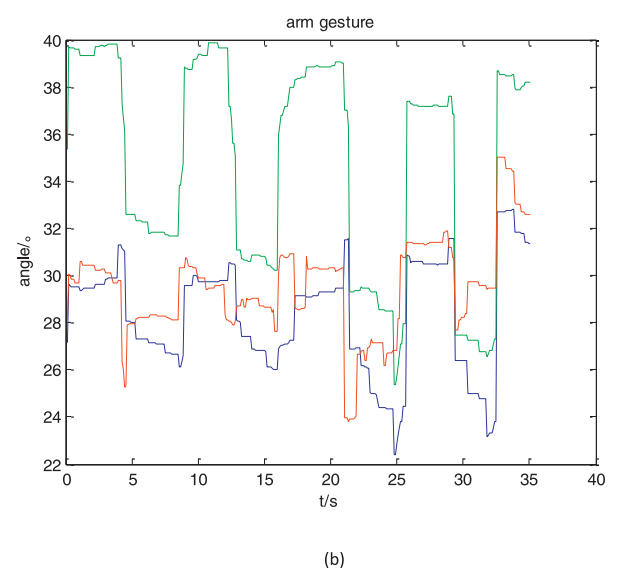
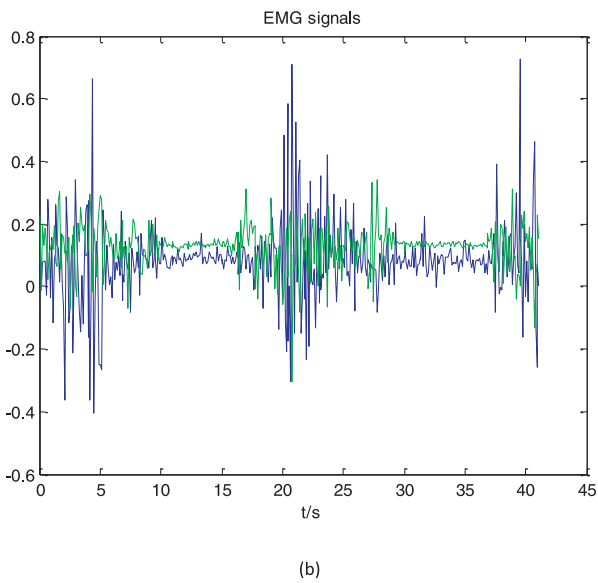
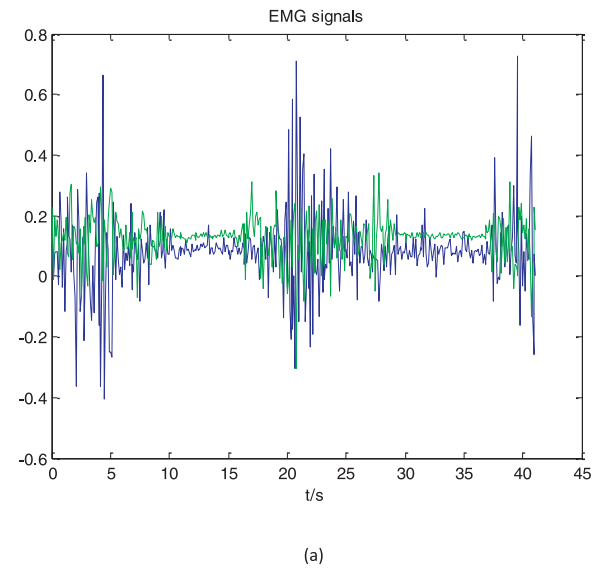
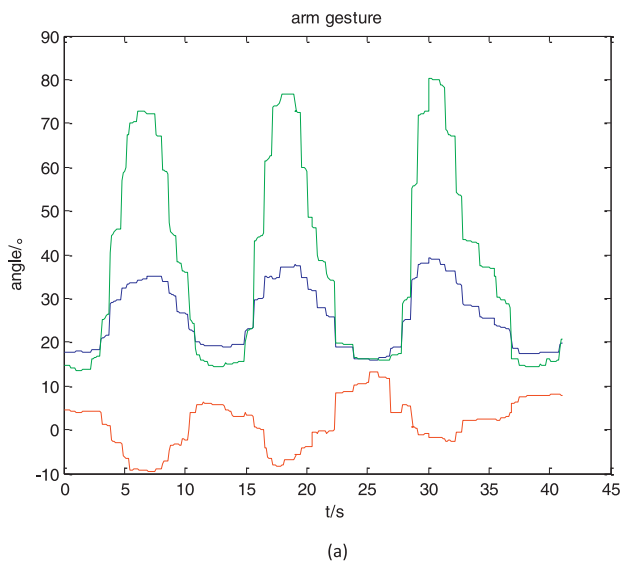


Fig. 13. Data and result of horizontal movement experiment.

Fig. 14. Data and result of vertical movement experiment.

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